WORKED SOLUTION

STEP I 2018, Question 3

Pure • Angles and Hyperbolas

STEP I 2018, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q3

The key idea

Both angles are tied to the coordinates of P through their tangents: the slope of RP gives $\tan\alpha$ and the slope of SP gives $\tan\beta$. The condition $\beta = 2\alpha$ then becomes a double-angle identity for tan, and clearing fractions gives the curve. Part (ii) runs the same algebra backwards. The catch is that knowing $\tan\beta = \tan 2\alpha$ only fixes β up to a multiple of π, so we have to track which quadrant 2α is in.

Solution

Setting up the angles

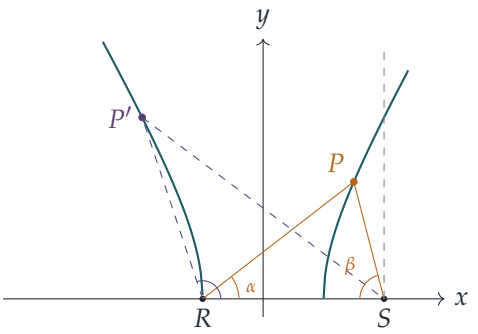
Since $y > 0$, the angle α at R is measured from the positive x-direction (along RS) round to RP, so $0 < \alpha < \pi$, and

$$\tan\alpha = \frac{y}{x+a} \qquad (x \neq -a).$$

This holds whether P is to the right of R (acute α, positive tangent) or to the left (obtuse α, negative tangent). At S, the angle β is measured from the negative x-direction (along SR) to SP. Because $x < 2a$, the point P is to the left of S, so β is acute and

$$\tan\beta = \frac{y}{2a-x} > 0, \qquad 0 < \beta < \tfrac{\pi}{2}.$$

Write $t = \tan\alpha$ when $x \neq -a$.



The curve $y^2 = 3(x^2 - a^2)$ with $y > 0$, $x < 2a$ (drawn for $a = 1$) has two pieces: an arc of the right branch with $a < x < 2a$, and the whole upper left branch $x < -a$. A point P' on the left branch makes α obtuse.

Part (i)

Suppose $\beta = 2\alpha$. Since $\beta < \frac{\pi}{2}$, we have $0 < \alpha < \frac{\pi}{4}$, so $x > -a$, $0 < t < 1$, and the double-angle formula applies with $1 - t^2 \neq 0$:

$$\frac{y}{2a - x} = \tan 2\alpha = \frac{2t}{1 - t^2} = \frac{2y/(x+a)}{1 - y^2/(x+a)^2} = \frac{2y(x+a)}{(x+a)^2 - y^2}.$$

Cancel y (allowed, as $y > 0$) and cross-multiply:

$$\begin{aligned} &(x+a)^2 - y^2 = 2(x+a)(2a - x) \\ \iff\quad &y^2 = (x+a)\big[(x+a) - 2(2a - x)\big] = (x+a)(3x - 3a) = 3(x^2 - a^2). \end{aligned}$$

So P lies on $y^2 = 3(x^2 - a^2)$, as required.

Part (ii)

Now suppose P lies on the curve, with $y > 0$ and $x < 2a$. Then $x^2 - a^2 = \frac{1}{3}y^2 > 0$, so $|x| > a$: either $a < x < 2a$ or $x < -a$. In particular $x \neq -a$, so $t = \tan\alpha$ is defined, and $t \neq 0$.

The steps of part (i) reverse. The curve's equation is equivalent to $(x+a)^2 - y^2 = 2(x+a)(2a - x)$. Dividing by $(x+a)^2$ and using $2a - x = y/\tan\beta$,

$$1 - t^2 = \frac{2(2a - x)}{x + a} = \frac{2y}{(x+a)\tan\beta} = \frac{2t}{\tan\beta}, \qquad \text{so} \qquad (1 - t^2)\tan\beta = 2t.$$

If $t^2 = 1$, this would give $t = 0$, which is impossible; so $t^2 \neq 1$ and

$$\tan\beta = \frac{2t}{1 - t^2} = \tan 2\alpha.$$

Ask yourself Does $\tan\beta = \tan 2\alpha$ mean $\beta = 2\alpha$?

Not necessarily: it means β and 2α differ by a multiple of π. We use what we know about the sizes of the angles, splitting into the two pieces of the curve.

Case $a < x < 2a$. Here $x + a > 0$, so $0 < \alpha < \frac{\pi}{2}$ and $0 < 2\alpha < \pi$. Since $\tan 2\alpha = \tan\beta > 0$, 2α lies in $(0, \frac{\pi}{2})$, the same interval as β. As tan is one-to-one on $(0, \frac{\pi}{2})$,

$$\beta = 2\alpha.$$

Case $x < -a$. Here $x + a < 0$, so $\frac{\pi}{2} < \alpha < \pi$ and $\pi < 2\alpha < 2\pi$. Since $\tan 2\alpha > 0$, $2\alpha \in (\pi, \frac{3\pi}{2})$, so $2\alpha - \pi \in (0, \frac{\pi}{2})$ and $\tan(2\alpha - \pi) = \tan 2\alpha = \tan\beta$. As before, tan is one-to-one on $(0, \frac{\pi}{2})$, so

$$\beta = 2\alpha - \pi.$$

So the possible relationships are $\beta = 2\alpha$ (when P is on the right-hand branch) and $\beta = 2\alpha - \pi$ (when P is on the left-hand branch). Both occur, since every point of each piece satisfies the conditions.

Checks

- Take $a = 1$ and $P = (-2, 3)$, which is on the curve since $3(4 - 1) = 9$. Then $\tan\alpha = 3/(-1) = -3$, so $\alpha \approx 108.43°$, and $\tan\beta = \frac{3}{4}$, so $\beta \approx 36.87°$. Indeed $2\alpha - 180° \approx 36.87°$.
- As $x \to 2a$ on the right branch, $y \to 3a$, so $\tan\alpha \to 1$ and $\tan\beta \to \infty$: $\alpha \to \frac{\pi}{4}$ and $\beta \to \frac{\pi}{2} = 2\alpha$.
- Far out on the left branch, P approaches the asymptote $y = -\sqrt{3}\,x$, so $\alpha \to \frac{2\pi}{3}$ and $\beta \to \frac{\pi}{3} = 2\alpha - \pi$.

A computer check of both relationships at many random points on each branch, for random a, also agreed.

Common slips

- Writing $\tan\alpha = y/|x + a|$, or assuming α is acute. On the left branch α is obtuse and $\tan\alpha < 0$.
- Concluding $\beta = 2\alpha$ from $\tan\beta = \tan 2\alpha$ without checking the ranges of the angles. This misses the second relationship.
- Forgetting that $x < 2a$ and $y > 0$ cut the curve down to two pieces, and that $|x| > a$ is needed to rule out $x = -a$.

Beyond STEP

The curve is a hyperbola with eccentricity 2, focus S and directrix $x = \frac{a}{2}$. The property in part (i) gives the classical way to trisect an angle using a hyperbola (known to Pappus), which gets round the impossibility of trisection with ruler and compasses alone. The impossibility proof itself is a highlight of a first course in field theory. Allenby's *Rings, Fields and Groups* covers it, and Needham's *Visual Complex Analysis* is a lovely companion on the geometry of angles and conics.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.